

Tabeeb Rahman

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EDUCATION

University of Bath

Bath, UK

BSc Mathematics & Physics (Predicted First)

Sep 2023 — May 2027

- Relevant Coursework: Advanced Data Analysis (90%), scientific computing (75%), Advanced Linear Algebra (66%), Thesis on Next Gen Spiking and Bayesian NN architectures, for: time series analysis, unsupervised deep learning.

A-levels: AAAB (Mathematics, Physics, Chemistry, Further Mathematics) from **Newham Collegiate Sixth Form**.

WORK EXPERIENCE

Junior Researcher and Data Analyst

Jun 2022 — Sep 2023

University College London & Zurich University

UCL, London

- Analysed attosecond particle physics experimental data, discovering new high energy electron behaviour.
- Contributed analysis to a peer-reviewed publication with international PhD collaborators.

Project Lead

Mar 2021 — Apr 2022

J.P Morgan Chase & Imperial College

Science Museum, London

- 2nd place in the national School's Challenge.
- designed and presented a vertical biofuel harvester to a panel of VC executives and ICL Engineers.

Engineering Lead

September 2023 — Apr 2024

Team Bath Drones

Univorsity of Bath, Bath

- 2nd place in the Institute of mechanical engineering European division UAV competition.
- Designed thermoelectric actuators, decreasing payload mechanism weight by 40%

PROJECTS

Developer, 3D Rigid Body Physics Engine

2019 — Present

- Explored and implemented advanced Geometry preserving numerical ODE solvers such as symplectic-euler.
- Used LCP methods, and quadratic programming for constrained resting contact complementarity problem, to efficiently compute collision impulse, Octrees, and implemented Phong shading.

Developer, Miscellaneous ML

2023 — Present

- Programmed CNN from First principles for image classification with OpenCL.
- Unsupervised Bayesian learning on SNNs, mitigating drift-variability issues in Next-Gen Memristive CIM AI accelerators.
- Work on SVMs leveraging Quadratic Programming and Zero-shot NLI, for natural language hypothesis testing.

Developer, Miscelaneous Full Stack Projects

2023 — Present

- Created dynamic web apps with React.js, demonstrating authentication and user access controls,
- PostgreSQL database integration with Prisma ORM, Cached with Redis and implemented Kubernetes HPA to deliver scalable enterprise E commerce, Content sharing and Business telemetry visualisation applications.

EXTRACURRICULAR ACTIVITIES

Mentorship & Outreach

2021 — Present

- Mentoring students from disadvantaged backgrounds, providing academic and interview support.
- Helped one Mentee secure a place at Imperial College London's medicine programme.

Student Leadership

2023 — Present

- Cohort leader at the University of Bath for the Mathematics and Physics department, via competitive selection.

SKILLS

- **Programming Languages:** Python, C/C++, Java/Javascript, Rust, SQL, MatLab.
- **Technologies:** OpenGL, TensorFlow, Nginx, PostgreSQL, Next.js, React.js, Node.js, Redis, Kubernetes, Prisma, PyTorch